



Sub-Theme : Climate Change Mitigation and Environmental Protection



TITLE:

**“BIOVOLT: ENGINEERING ANTHOCYANIN AND CHLOROPHYLL
BASED DYE-SENSITIZED SOLAR CELLS (DSSCS) UTILIZING
FRUIT AND VEGETABLE WASTE FOR CLIMATE CHANGE
MITIGATION AND URBAN ENVIRONMENTAL PROTECTION”**

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I. INTRODUCTION

The global climate crisis demands a transformations toward regenerative technological ecosystems. Indonesia faces a environmental challenge on systemic organic waste ineffective administration and severe energy vulnerability from fossil fuel dependency. Annual food loss and waste (FLW) in Indonesia reaches 23–48 million tons, dominated by fruit and vegetable waste. In open dumpsites, this biomass undergoes uncontrolled anaerobic decomposition, releasing hazardous leachate and methane (CH_4), a pollutant with a Global Warming Potential 28 times greater than CO_2 . Concurrently, the national power grid remains over 60% dependent on coal fired generation, driving massive carbon emissions and grid vulnerability. Clean energy alternatives are urgently needed to resolve both sanitation and energy deficits.

To address these challenges, BioVolt is an engineered Dye-Sensitized Solar Cell (DSSC) technology that transforms organic waste into clean power. By extract natural photo sensitizers (anthocyanins and chlorophyll) from discarded produce, BioVolt convert organics waste into functional photovoltaics components, preventing landfill methane emissions while harvesting indoor and diffuse light power ($15 - 30 \text{ W}_p/\text{m}^2$). This inovationn directly advances the United Nations Sustainable Development Goals, specifically SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), SDG 9 (Industry, Innovation, and Infrastructure), SDG 7 (Affordable and Clean Energy), and SDG 17 (Partnerships for the Goals).



Fig.1 Relation between SDGs and Biovolt

II. CONTENT AND DISCUSSION

A. Technical Architecture, Photophysics, and Component Manufacturing

BioVolt re-engineers conventional single crystalline silicon solar cells by utilizing photon-induced electrochemical charge transfer mechanisms. The functional manufacturing process and underlying scientific principles are integrated through three stages:

1. Photon Excitation and Pigment Formulation

Fruit and vegetable waste (such as dragon fruit peels and rejected spinach) is collected from central markets, homogenized in an ethanol-citric acid solvent, and centrifuged for 15 minutes at 4000 RPM. Visible light photons absorbed by natural anthocyanins and chlorophyll excite electrons from the Highest Occupied Molecular Orbital (HOMO) to the Lowest Unoccupied Molecular Orbital (LUMO). A blended pigments calibration algorithm optimizes the optical absorption coefficient (α) across the 400-700 nm spectrum, maintaining stable shorts circuit current density (J_{sc}) despite seasonal variations in waste feedstock composition.



2. Photoanode Deposition and Esterification Binding

Fluorine doped Tin Oxide (FTO) conductive glass substrates are coated with an anatase Titanium Dioxide (TiO_2 , ~ 20 nm) paste using the doctor blade technique and calcined at $450^\circ\text{C} - 500^\circ\text{C}$ for 45 minutes to form a porous mesoporous nanostructure. Immersing the photoanode in the pigment extract for 18 hours forms stable esterification bonds (covalent coordination) between the carboxyl/hydroxyl groups of the dye and Titanium atoms. Because the LUMO level of the pigment lies above the Conduction Band (CB) of TiO_2 (~ -4.2 eV), ultrafast electron injection occurs, minimizing interfacial charge transport resistance.

3. Low Cost Cathode Fabrication and Electrolyte Regeneration

The cathode is produced by depositing low cost amorphous carbon (graphite soot derived from biomass pyrolysis) onto FTO glass, eliminating expensive, high carbon footprint platinum (Pt) catalysts. The cells are encapsulated with Surlyn sealant at 120°C and injected with an iodide/triiodide (I^-/I_3^-) redox couple electrolyte to continuously regenerate oxidized dye molecules.

B. Photovoltaic Performance and Power Output Generation

Under standard laboratory testing conditions (Air Mass 1.5G, $100 \text{ mW}/\text{cm}^2$), BioVolt natural dye-sensitized solar cells exhibit the following electrical characteristics:

Open Circuit Voltage (V_{oc}): $\sim 0.55 \text{ V} - 0.65 \text{ V}$ per single cell.

Short Circuit Current Density (J_{sc}): $\sim 1.5 \text{ mA}/\text{cm}^2 - 3.5 \text{ mA}/\text{cm}^2$.

Fill Factor (FF): $0.55 - 0.68$.

Power Conversion Efficiency (η): Ranging between 1.5% and 3.2%.

Power Output Estimation per BIPV Module:

When implemented on architectural glass modules scaled at 1 m^2 , BioVolt generates a peak output of $15 - 30$ Watt-peak (W_p). Unlike rigid silicon panels whose efficiency drops drastically under overcast weather or artificial indoor lighting, BioVolt maintains robust energy harvesting under diffuse and low intensity light ($200 - 1000$ lux), making it optimal for vertical Building Integrated Photovoltaics (BIPV) in urban high rises.

C. Sustainability Analysis and Climate Change Mitigation

1. Methane Emission Abatement

Diverting municipal fruit and vegetable waste prevents uncontrolled anaerobic decomposition in landfills. With methane possessing a Global Warming Potential (GWP_{100}) 28 times higher than CO_2 , each ton of organic waste converted into BioVolt photosensitizers abates approximately $0.8 - 1.2$ tons of CO_2 equivalent emissions.

2. Circular Economy Without Toxic or Rare Metals

BioVolt eliminates synthetic dyes containing toxic Ruthenium (Ru) complexes and removes dependence on mined Platinum (Pt), utilizing abundant agricultural waste and biomass derived carbon instead.



3. Degradation Management via Re-Sensitization

Resolve natural dye photobleaching under prolonged UV exposure, BioVolt incorporates micro fluidic channels within the panel assembly. An automated dye replenishment service flushes and refills fresh organic extract every 2–3 years, extending the operational life of the glass substrate without requiring total panel replacement.

D. Primary Market Survey Analysis: Consumer Acceptance & Purchase Intent

To validate market acceptance and commercialization feasibility, primary survey data was collected from target end-users, confirming problem urgency and market demand:

1. Problem Validation

- o Organic Waste Generation

79.7% of respondents reported a high frequency (scores 3 and 4) of generating daily organic waste (fruit peels, vegetable scraps, kitchen waste), while 48.4% explicitly noted that current waste management practices remain ineffective.

- o Energy Vulnerability

18.7% of respondents confirmed experiencing power outages (scores 3 and 4), reinforcing the need for decentralized renewable energy solutions.

2. Consumer Interest & Purchase Intent

- o Concept Acceptance

75% of respondents expressed strong interest in the core concept of converting organic waste into renewable solar energy via BioVolt.

- o Purchase Intention

When asked if they would adopt BioVolt upon market availability, 96.9% of respondents (62 out of 64 participants) indicated direct interest, providing empirical market validation.

3. Consumer Advocacy (Word-of-Mouth Potential)

Respondents demonstrated a high willingness to recommend BioVolt to peers, scoring an average of 3.47 out of 4.00 (with 60.9% giving the maximum rating of 4.00), demonstrating strong organic marketing potential.

E. Techno Economic Feasibility and Circular Business Model

BioVolt presents a scalable economic model for real world. Through partnerships with municipal market authorities, BioVolt secures its primary feedstock at zero cost (IDR 0), insulating manufacturing from raw material price volatility and reducing Cost of Goods Sold (COGS).

Financial evaluation for a standard batch capacity of 100 units/month confirms a 42.5% gross profit margin and strong capital return. Research & Development (R&D) and Intellectual Property (IP) maintenance costs are embedded into operational expenses. Detailed calculations are presented in Appendix A.

F. Strategic Marketing Framework (STP & 4Ps)

- Segmentation

High density metropolitan urban centers and central traditional market hubs.



- **Targeting**
B2B eco conscious real estate developers, green architectural firms, corporate property managers, and EPC solar contractors.
- **Positioning**
Premium, because it integrated semi transparent architectural glass converting waste into clean energy.
- **Marketing Mix (4Ps):**
 - Product: Semi transparent colorful BIPV glass panels with an integrated dye replenishment maintenance scheme.
 - Price: Competitive pricing at IDR 800,000 / unit.
 - Place: Direct distribution to green developers and market micro grid installations.
 - Promotion: B2B engagement, demonstration projects, and ESG carbon reduction reporting for corporate partners.

G. Scaling and Corporate Exit Strategy

1. **Merger & Acquisition (M&A)**
Technology licensing or acquisition by international glass manufacturer or renewable energy conglomerates looking to integrate organic BIPV layers.
2. **Joint Venture (JV) & Initial Public Offering (IPO)**
Partnerships with state owned energy enterprises for market micro grid, building toward an IPO on the Indonesia Stock Exchange.

III. CONCLUSION AND RECOMMENDATIONS

BioVolt successfully demonstrates the technical, environmental, and economic viability of engineering municipal fruit and vegetable waste into high performance Dye Sensitized Solar Cell (DSSC) technology. By utilizing natural anthocyanin and chlorophyll dyes bonded via esterification to mesoporous TiO_2 , alongside low cost biomass derived carbon cathodes, BioVolt delivers an impact climate solution it mitigates landfill methane (CH_4) emissions while replacing fossil fuel based electricity with continuous indoor and diffuse power generation ($15 - 30 \text{ W}_p/\text{m}^2$). This innovation advances SDGs 7, 9, 12, 13, and 17. The innovation's strong techno economics profile is evidenced by a Cost of Goods Sold (COGS) of IDR 460,000/unit, an R/C ratio of 1.74, and a monthly ROI of 73.9%. Furthermore, primary market survey data (Enclosure E) validates overwhelming consumer acceptance, with 79.7% of respondents confirming severe organic waste issues, 75% expressing strong interest, and 96.9% demonstrating explicit purchase intent. To accelerate market and long term implementation, future technical efforts must focus on integrating natural cross linking agents to extend dye replenishment intervals, supported by policy frameworks such as feed in tariffs, tax credits, and ESG carbon offset schemes for waste to energy BIPV adoption.



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APPENDICES

A. Techno-Economic & Financial Analysis

Evaluation for a batch capacity of 100 units/month:

Fixed Cost (FC): IDR 16,500,000 / month

Variable Cost (VC): IDR 24,500,000 / month

R&D & IP Cost (RC): IDR 5,000,000 / month

Total Cost (TC): $TC = FC + VC + RC = \text{IDR } 46,000,000/\text{month}$

Cost of Goods Sold (COGS / HPP): $HPP = \frac{TC}{100 \text{ units}} = \text{IDR } 460,000/\text{unit}$

Revenue & Profitability Indicators:

Selling Price: IDR 800,000 / unit (Gross profit margin: 42.5% or IDR 340,000/unit)

Gross Revenue: $100 \text{ units} \times \text{IDR } 800,000 = \text{IDR } 80,000,000/\text{month}$

Revenue-to-Cost (R/C) Ratio: $\frac{\text{IDR } 80,000,000}{\text{IDR } 46,000,000} = 1.74$

Monthly ROI: $\frac{\text{IDR } 34,000,000 \text{ Net Profit}}{\text{IDR } 46,000,000} \times 100\% = 73.9\%$

B. PESTLE Analysis Matrix

Dimension	Analysis & Climate Context
Political	Strong alignment with national net-zero targets and Visi Indonesia Emas 2045.
Economic	Zero-cost raw material shields operations from commodity price volatility. High monthly ROI (73.9%).
Social	Rising public awareness regarding urban sanitation and green building aesthetics.
Technological	Superior low-light/indoor performance over conventional silicon panels.
Legal	Compliance with waste governance laws and securing IP patent protection.
Environmental	Direct prevention of methane (CH ₄) emissions from municipal waste decomposition.

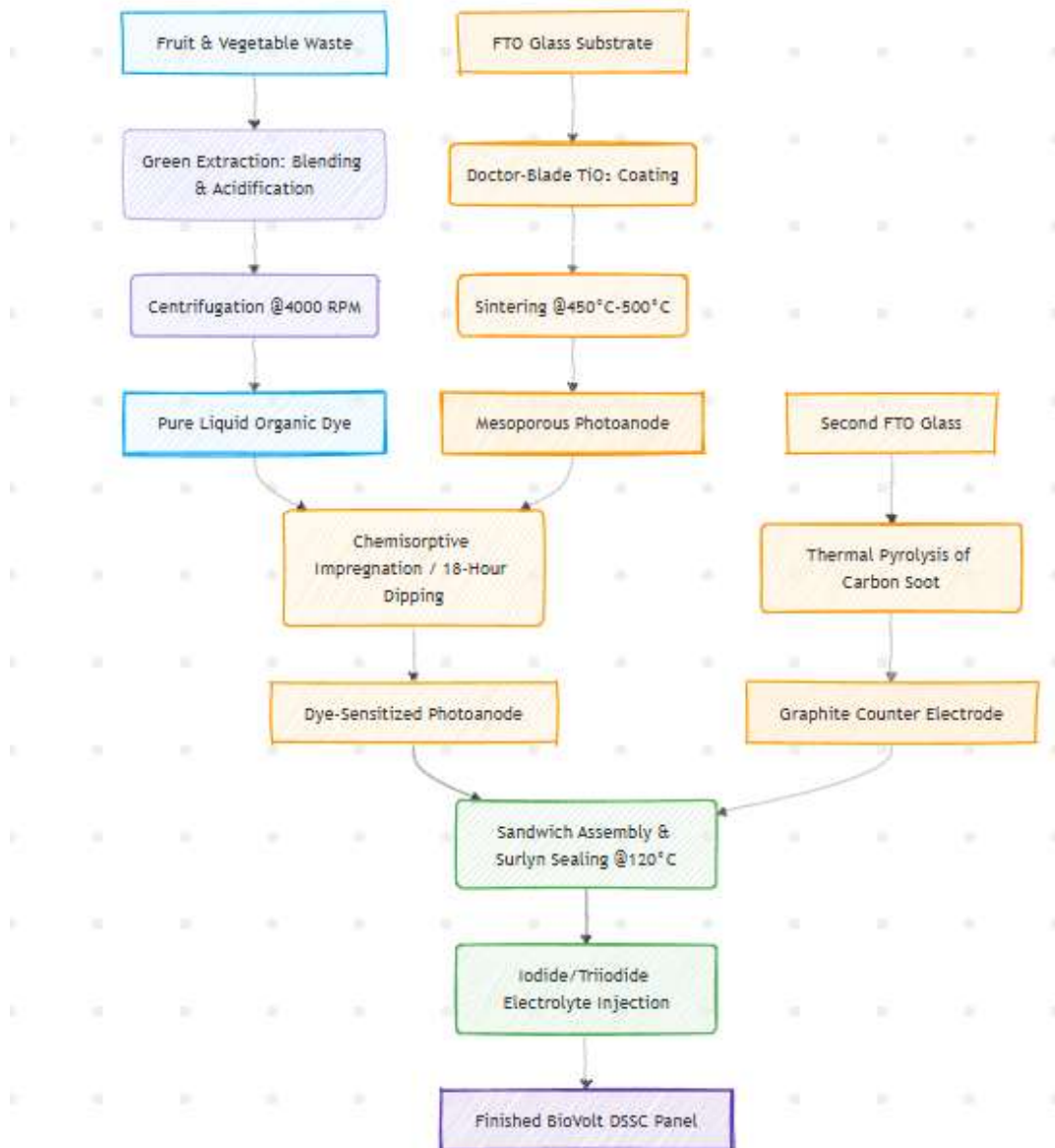
C. SWOT Analysis Matrix

STRENGTHS Zero-cost organic dyes, high ROI (73.9%); strong low-light performance.	WEAKNESSES Lower baseline conversion efficiency compared to silicon, UV dye photobleaching over time.
OPPORTUNITIES Growing demand for ESG/Greenship certifications and carbon tax implementation.	THREATS Aggressive price drops in standard silicon panels, supply chain delays for FTO glass.

D. Quantitative Environmental Impact Evaluation

Environmental Metric	Direct Impact per 1,000 BioVolt Panels Installed	Environmental Protection Outcome
Organic Waste Diversion	≈ 120 Tons / Year diverted from municipal dumpsites	Prevents groundwater leachate contamination and eliminates bad odor in market areas.
Methane Abatement	≈ 8.4 Tons CH ₄ / Year prevented from decaying waste	Directly reduces high-GWP greenhouse gas emissions at the regional level.
Carbon Offset (Electricity)	≈ 145 Tons CO ₂ Equivalent / Year replaced from coal grid	Accelerates clean energy transition and lowers corporate carbon intensity.

E. Technical Flow of Architecture and Component Manufacturing





F. Primary Market Survey Data & Raw Responses

This appendix of the complete dataset, survey instrument, and raw response tallies collected from target consumers regarding organic waste management issues and market demand for BioVolt.

Primary Raw Dataset: Available via Google Sheets at

<https://docs.google.com/spreadsheets/d/14wJdhoPSeAdh3JoUSvyGh5nlqmqzVuiGiLtR1Ff8E7KI/edit?usp=sharing>

Summary of Survey Findings ($N = 64$):

- Organic Waste Concern: 79.7% ($n = 51$) of respondents confirmed severe local organic waste management issues.
- Concept Interest: 75.0% ($n = 48$) expressed strong interest in adopting BioVolt technology.
- Purchase Intent: 96.9% ($n = 62$) demonstrated explicit purchase intent upon market release.